VANDERBILT HEALTH | Specialty Pharmacy

Optimizing Maribavir Management: The Role of Health System Specialty Pharmacies in Access, Monitoring, and Waste Reduction

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HIGHLIGHTS

- Pharmacists facilitated access to maribavir by obtaining timely insurance prior authorizations (median = 1 day) and intervened to reduce 28 (30%) fills across 94 dispenses to avoid waste.
- Maribavir quantity was reduced for 10 dispenses during the final treatment course, based on CMV levels and time until next lab appointment, which resulted in $119,517 - $149,396 in avoided costs.

Purpose

Maribavir, which treats cytomegalovirus (CMV) infection in post-transplant recipients who are refractory to first-line treatments, is a high cost, limited distribution medication, that requires frequent lab monitoring to assess for efficacy.

This study evaluated the specialty pharmacist's role in medication access, treatment monitoring, and reducing maribavir waste.

Study Design

Single-center, retrospective cohort analysis of data collected from an electronic medical record and specialty pharmacy management system.

Patients were included if they were prescribed maribavir from 4/1/2022 to 12/31/2023. Patients were followed from the time of maribavir initiation to four months after therapy discontinuation.

Methods

Patient Population

Patients who initiated maribavir for post-transplant CMV infection/disease that is refractory to treatment (with or without genotypic resistance) with ganciclovir, valganciclovir, cidofovir, or foscarnet

$ Calculating Waste Avoided

Medication costs calculated using:

- Wholesale acquisition cost (WAC)
- Average wholesale price (AWP)
- AWP minus 20%

Cost avoidance calculation:

14-day supply of maribavir not dispensed due to PharmD intervention X AWP / AWP minus 20% / WAC

Figure 1. Study Attrition

Prescribed maribavir
All patients initially received a 28-day supply of maribavir. Five patients were required to repeat treatment with maribavir due to CMV reactivation.
(n=27 patients with 34 instances of maribavir use)

→ Maribavir filled at Vanderbilt Specialty Pharmacy
(n=23 with 94 dispenses)

→ Maribavir filled at external specialty pharmacy or through PAP
(n=4)

Results

Figure 2. Patient Demographics (n=27)

Gender
33% Female (n=9)
67% Male (n=18)

Age
Median
62 years
IQR 52,65

Race
15% Black (n=4)
85% White (n= 23)

Pharmacy Insurance
52% Commercial (n=14)
44% Medicare (n=12)

Transplant Type[a]
11% Heart (n=3)
33% Lung (n=9)
37% Kidney (n=10)
15% Liver (n=4)
[a]Multi-organ n=1, 4%

CMV Status
100% Donor CMV Status
78% Recipient CMV Status

Indication
Resistance to prior treatment 44%
56% Refractory to prior treatment

Figure 3. Maribavir Access and Monitoring: Pharmacist's Workflow

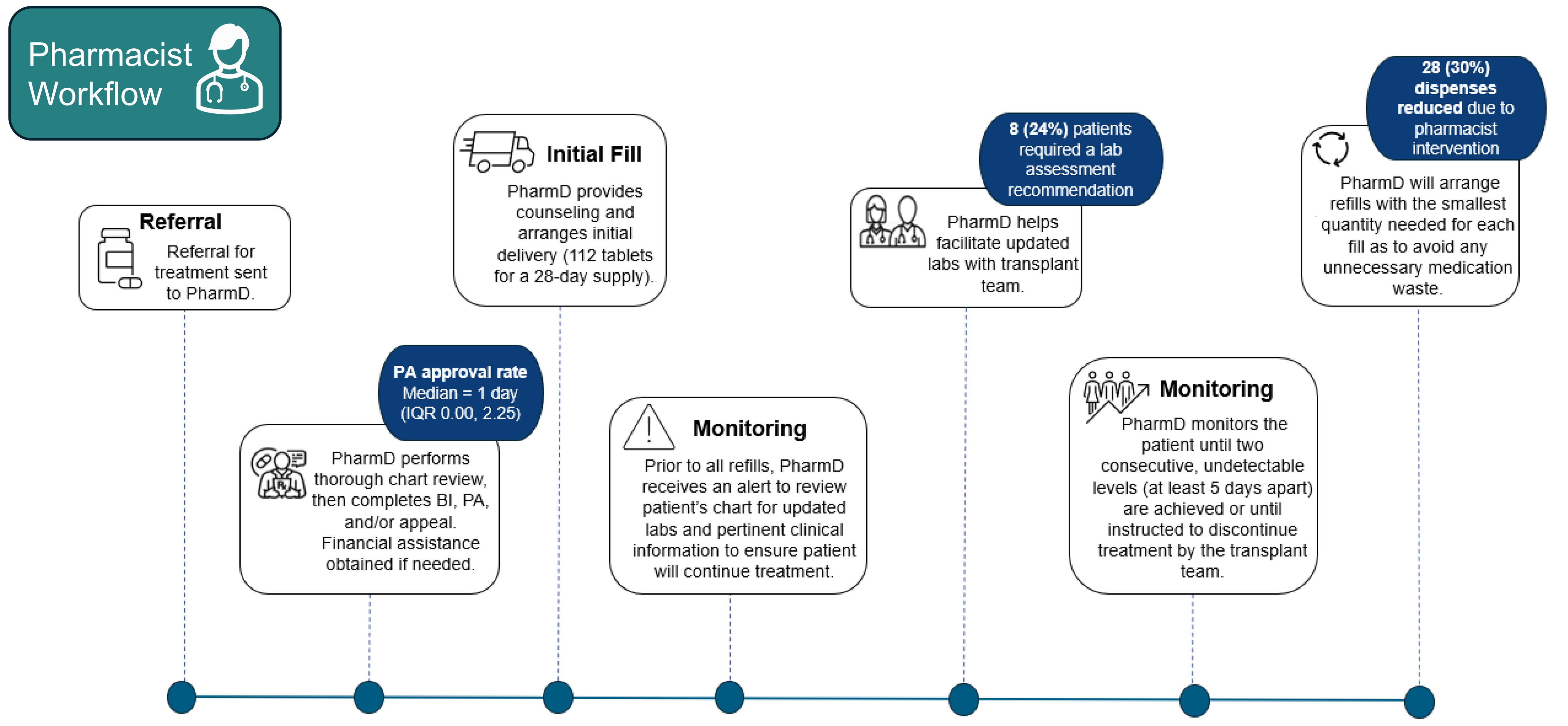


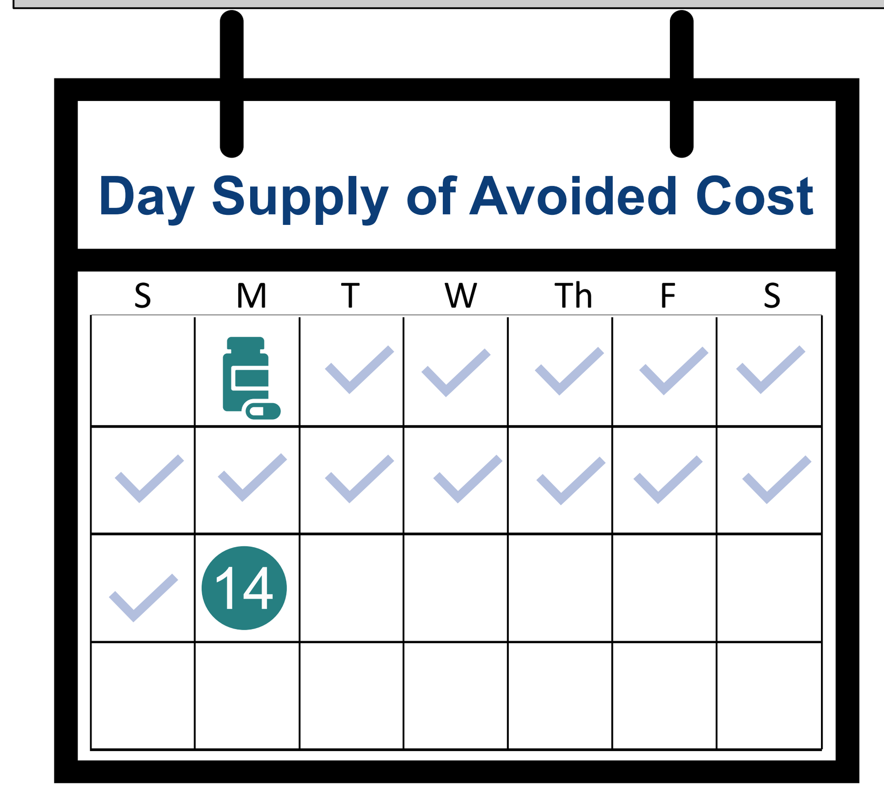
Figure 4. Waste Avoidance Intervention

Pharmacist intervention to reduce quantity filled from 28-day supply to 14-day supply depending on CMV level and time until next lab appointment
(n=28 dispenses)

CMV still detectable:
Pharmacist intervention **does not** lead to waste avoidance
(n=18 dispenses)

CMV undetectable:
Pharmacist intervention **leads** to waste avoidance during final treatment course
(n=10 dispenses)

Figure 5. Cost Avoided



	AWP-20%	WAC	AWP
Total Cost Avoided	$119,517.44	$124,499.20	$149,396.80
Cost Avoided per Fill	$11,951.74	$12,449.92	$14,939.68

Abbreviations: CMV = cytomegalovirus; PAP = patient assistance program; PharmD = pharmacist; BI = benefits investigation; PA = prior authorization; IQR = interquartile range

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